

NPTEL ETHICAL HACKING & NETWORK COMMUNICATIONS

WEEK 1 DETAILED CLASSROOM STUDY NOTES & EXAM PREPARATION NOTEBOOK

KEY CONCEPT TO REMEMBER: TEACHER'S NOTE & LEARNING METHODOLOGY

Welcome to Week 1 Study Notes! Cybersecurity and ethical hacking rest entirely on a rock-solid understanding of computer networking. In this notebook, every concept is taught from basic principles using a 4-step framework: **[1] What it is** → **[2] Why it is needed** → **[3] How it works step-by-step** → **[4] Real-world example/analogy**. Mathematical formulas, network delay calculations, IP header structures, and exam practice questions are fully worked out in detail. Study systematically and review the revision pages at the end before your NPTEL exam!

MODULE 1: LECTURE 01 — INTRODUCTION TO ETHICAL HACKING & PENETRATION TESTING

1.1 What is Ethical Hacking?

1. Simple Explanation: Imagine you own a high-security bank vault. Instead of waiting for a bank robber to try to break in and steal your money, you hire a trusted security professional to test your locks, security cameras, and guard responses by trying to break in themselves (with your permission). When they find a flaw, they tell you exactly how they broke in so you can fix it. That authorized security professional is an **ethical hacker**.

2. Why it is Needed: Modern digital infrastructure (websites, cloud databases, banking portals) is under constant scan by malicious actors. Organizations need a proactive defense strategy. Waiting for an attack leads to data theft, financial loss, and loss of public trust. Ethical hacking allows organizations to discover and patch software flaws *before* criminals exploit them.

3. Technical Definition: *Ethical Hacking* is defined as the authorized act of locating weaknesses and vulnerabilities in computer systems, applications, and network infrastructures by systematically mimicking the intent, techniques, and actions of malicious hackers.

4. How it Works Step-by-Step:

- 1. Authorization & Scope Definition:** Obtain explicit legal written consent from the target system owner detailing boundaries, permitted tools, and testing schedules.
- 2. Reconnaissance & Asset Mapping:** Gather public and network information to map out the target's attack surface (e.g., via port scanning).
- 3. Vulnerability Identification:** Analyze running services, open ports, and software versions for known security flaws.
- 4. Simulated Exploitation:** Safely execute controlled exploit scripts to demonstrate how an attacker could breach authentication or gain access.
- 5. Reporting & Remediation:** Document findings in a detailed report highlighting identified risks, proof-of-concept evidence, and concrete technical fixes.

REAL-WORLD EXAMPLE / ANALOGY: Bank Vault Security Guard Analogy

An ethical hacker acts like a hired security consultant testing a building's perimeter locks. A malicious hacker (cracker) is the burglar attempting to steal physical assets without permission.

EXAM POINT: LEGAL & ETHICAL BOUNDARIES

Ethical hacking operates under strict legal authorization. Operating without explicit written consent—even in good faith or to demonstrate a flaw—violates cybercrime laws. Universal prohibitions include: (1) Unconsented system access, (2) Installing worms/viruses, (3) Launching Denial of Service (DoS) attacks, and (4) Denying legitimate users access to network resources.

1.2 Core Terminology & Attack Vectors

Understanding cybersecurity requires mastering precise domain terminology:

Technical Term	Technical Definition	Security Significance & Context
Hacking	Broadly refers to applying technical expertise to manipulate, gain access to, or control computer systems.	Not inherently evil; refers to advanced technical skill set applied ethically for defense or maliciously.
Cracking	The specific act of breaching system security (software, hardware, or networks) without authorization.	A malicious action that ethical hackers simulate in educational/authorized environments to test resilience.
Penetration Testing	A formal, structured type of ethical hacking focused on breaking into a target to evaluate security posture.	Provides realistic assessments of organizational defenses and proves whether vulnerabilities are exploitable.
Spoofing	An attack where a perpetrator falsifies their identity, typically by forging source addresses in network packets.	Highlights the critical need for strict identity verification, cryptographic signing, and traffic inspection.
Denial of Service (DoS)	An attack designed to render a system or network resource unavailable by flooding it with excessive traffic.	Disrupts business continuity. Ethical hackers test capacity limits to ensure high availability.
Port Scanning	Reconnaissance process of sending probe packets to active ports (entry points) to discover running services.	Crucial initial step for mapping an attack surface and identifying vulnerable running software versions.
Backdoor	A hidden entry point into a system, often left by developers for debugging, that bypasses standard login.	Represents severe risk if unmanaged; allows persistent, unauthorized access bypassing passwords.
Trojan Horse	Malicious software disguised as legitimate code, or embedded in hardware during fabrication.	Gains initial entry under the guise of useful software; drops backdoors or steals data upon execution.
Software Exploitation	Leveraging known software bugs or unpatched flaws to execute arbitrary code or elevate privileges.	Primary target for pen testers; unpatched published vulnerabilities are easily exploited by script kiddies.
Script Kiddies	Inexperienced attackers who run pre-written scripts and automated tools downloaded online.	Baseline threat; lack deep knowledge but cause heavy damage to unpatched, misconfigured systems.
Tiger Box	A portable, self-contained laptop toolkit pre-configured with security OSs and penetration testing tools.	Essential mobile hardware toolkit used by security professionals for efficient field vulnerability assessments.

1.3 Penetration Testing Models (White, Black, and Gray Box)

Ethical hackers conduct penetration tests using three distinct information models depending on the client's testing goals:

Testing Model	Information Provided to Tester	Real-World Scenario Simulated	Primary Advantages & Trade-offs
White Box Model	Full information (network topology, IP schemes, source code, credentials, IT staff access).	Simulates an internal audit or a malicious insider with full network clearance.	Maximum thoroughness and efficiency; tests all internal code and architecture rapidly.
Black Box Model	Zero prior information. Tester only knows company name or domain name.	Simulates a blind external cyberattack from an outside malicious hacker.	Highly realistic external assessment; however, time-consuming during reconnaissance.
Gray Box Model	Partial information (e.g., user-level credentials, limited IP subnets, or partial documentation).	Simulates an attacker who breached perimeter defenses or a third-party partner.	Balances cost, time efficiency, and realistic attack simulation. Most common in enterprise testing.

1.4 Post-Access Actions & Legal Frameworks

Once an attacker or pen-tester breaches a system, several post-exploitation actions typically occur:

- **Log Modification:** Attackers alter or delete system logs (e.g., `/var/log/auth.log`) to erase digital footprints and evade detection.
- **Data Manipulation:** Theft of sensitive databases, alteration of financial records, or defacement of public websites.
- **Installing Persistent Backdoors:** Setting up hidden accounts or SSH keys to maintain access even if the main flaw is patched.

• **Lateral Movement:** Using the compromised host as a jump-point (pivot) to attack other systems deeper inside the internal LAN.

COMMON CONFUSION / MISTAKE: Script Kiddies vs. Experienced Testers

Script Kiddies copy-paste pre-written exploits without understanding the code. **Experienced Testers** write custom exploit code in Python, Perl, JavaScript, or C/C++, allowing them to bypass security filters and craft targeted payloads.

MODULE 2: LECTURE 02 — COMPUTER NETWORKING FUNDAMENTALS (PART I)

2.1 Why Networking is Essential for Ethical Hacking

A computer network is a communication system connecting diverse computing devices (hosts, servers, gadgets, equipment) enabling them to exchange data and share hardware/software resources.

Why Ethical Hackers Must Master Networking: You cannot secure or audit what you do not understand. Ethical hacking exploits mechanisms in network protocols, packet routing, and connection handshakes. Without a thorough grasp of how data travels across links, identifying vulnerabilities like packet sniffing, IP spoofing, or session hijacking is impossible.

2.2 Classification of Networks by Geographical Scale

Networks are categorized primarily by their physical and geographical span:

Network Scale	Geographical Coverage	Standard Technology & Speeds	Key Characteristics & Use Cases
Local Area Network (LAN)	Small geographical area: single room, office, building, or university campus.	Ethernet (IEEE 802.3). Speeds evolve from 10 Mbps to 10 Gbps, 40 Gbps, 100 Gbps+.	High-speed, low cost, privately owned infrastructure. Used for localized resource sharing.
Metropolitan Area Network (MAN)	City-wide scale (e.g., connecting multiple municipal offices across a city).	Fiber optic rings, Metro Ethernet.	Intermediary scale between LAN and WAN. Often managed by telecom consortiums.
Wide Area Network (WAN)	Large geographical areas spanning cities, countries, or whole continents.	Leased lines, Fiber backbones, Satellite links, Routers.	Connects distant LANs across public/commercial infrastructure. Higher cost, lower speeds than LANs.

2.3 Data Communication Techniques: Circuit vs. Packet Switching

When data moves from sender to receiver across intermediate nodes, networks use one of two fundamental transmission paradigms:

A. Circuit Switching (Traditional / Analog Approach)

In circuit switching, a dedicated physical communication path is established between sender and receiver before transmission begins.

The 3 Sequential Phases:

1. **Connection Establishment:** A path is set up through intermediate switches. Logical channels are reserved over physical links, guaranteeing fixed bandwidth (e.g., 64 kbps per voice call).
2. **Data Transfer:** Data flows at maximum link speed continuously without queuing or headers.
3. **Connection Termination:** Dedicated links and switch resources are released for other callers.

REAL-WORLD EXAMPLE / ANALOGY: Traditional Landline Telephone Analogy

When you make a traditional landline phone call, copper lines and exchange switches reserve a dedicated physical circuit exclusively for your call until you hang up. Even if you stay silent, no one else can use that bandwidth.

B. Packet Switching (Modern / Digital Approach)

Instead of reserving dedicated lines, messages are segmented into smaller units called **packets**. Each packet receives a header containing source/destination addresses and sequence numbers, then travels independently across shared network links.

Key Mechanisms of Packet Switching:

- **Shared Links:** Communication lines are shared dynamically among thousands of users.
- **Store and Forward:** Intermediate routers receive a full packet into a memory buffer, inspect its header, and forward it to the next link. This enables: (1) *Link Utilization* during bursty gaps, (2) *Data Rate Conversion* (buffering fast incoming links feeding slower outgoing links), and (3) *Packet Prioritization* (high-priority packets jumped ahead in queue).

• **Routing Tables:** Data tables in routers mapping destination subnets to specific outgoing interfaces.

EXAM POINT: WHY CIRCUIT SWITCHING FAILS FOR COMPUTER DATA

Computer data is inherently '**bursty**'—a user downloads a web page (high data burst) and then reads it for 2 minutes (zero data flow). Reserving a dedicated circuit wastes 99% of bandwidth during silent periods. Packet switching allows thousands of users to share the link during idle gaps!

2.4 Hybrid Approach: Virtual Circuits

What it is: Virtual Circuit switching is a *connection-oriented* packet switching approach that combines elements of both paradigms.

How it Works:

1. **Connection Setup:** A setup packet traverses the network to determine a fixed logical path. Intermediate routers log this route and assign a unique **Virtual Circuit Number (VCN)**.
2. **Data Transfer:** Packets carry only the lightweight VCN in their headers (not full 32-bit IP addresses). Routers forward packets based on VCN routing tables using store-and-forward.
3. **Shared Infrastructure:** Physical links remain shared among multiple virtual circuits; bandwidth is not dedicated.

COMMON CONFUSION / MISTAKE: CRITICAL DRAWBACK OF VIRTUAL CIRCUITS

Because the path is fixed during setup, **Virtual Circuits cannot dynamically adapt** to network congestion or link failures. If an intermediate link fails, all packets on that virtual circuit drop until a new connection setup phase is executed!

MODULE 3: LECTURE 03 — NETWORKING FUNDAMENTALS (PART II): DATAGRAMS, DELAYS & OSI

3.1 The Datagram Approach to Packet Switching

Concept: The datagram approach is the fundamental packet switching method used across the global Internet. Each packet is treated as an entirely *independent entity* carrying full source and destination IP addresses. No prior path or virtual circuit is established.

REAL-WORLD EXAMPLE / ANALOGY: Postal Service Letter Analogy

Sending datagrams is exactly like dropping 5 letters into a mailbox addressed to the same recipient. Each letter travels independently on different mail trucks or flights. They may arrive at different times or out of order, but the recipient uses page numbers to reassemble the message.

Operational Characteristics:

- **No Call Setup Delay:** Transmission begins immediately without connection handshakes.
- **Dynamic Routing Decisions:** Every intermediate router inspects incoming datagram headers individually and chooses the optimal outgoing link based on real-time traffic conditions.
- **Resilience & Flexibility:** If a router or link fails mid-transmission, subsequent datagrams automatically reroute around the failure.

EXAM POINT: DATAGRAM ANOMALIES TO MEMORIZE FOR EXAM

Because datagrams take independent paths, three anomalies occur: (1) **Out-of-Order Delivery** (packets taking faster paths arrive before earlier packets), (2) **Packet Loss** (router buffer overflows discard packets), and (3) **Duplicate Packets** (sender retransmits when acknowledgments are delayed). Upper layers (TCP) must handle reordering and retransmission!

3.2 Components of Network Delay & Mathematical Formulas

Network performance and packet latency are governed by four additive delay components:

1. Propagation Delay (T_{prop}): Time required for a signal's electromagnetic wave to travel across the physical medium from sender to receiver.

Formula: $T_{prop} = d / s$

where **d** = physical distance (meters), **s** = propagation speed of light in medium ($\approx 2 \times 10^8$ m/s in copper/fiber, 3×10^8 m/s in air).

2. Transmission Time (T_{trans}): Time required to push all bits of a packet onto the physical communication link.

Formula: $T_{trans} = L / R$

where **L** = packet length in bits, **R** = link bandwidth/transmission rate in bits per second (bps).

3. Processing Delay (T_{proc}): Microseconds spent by intermediate routers examining headers, verifying checksums, and consulting routing tables.

4. Queueing Delay (T_queue): Time a packet spends waiting in router memory buffers before transmission (depends on network congestion).

KEY CONCEPT TO REMEMBER: TOTAL NETWORK DELAY EQUATION

Total Latency (T_total) = T_prop + T_trans + T_proc + T_queue

Note: Propagation delay depends on distance and physical medium. Transmission time depends on packet size and bandwidth. They are completely independent!

3.3 Master Comparison: Circuit vs. Virtual Circuit vs. Datagram

A classic NPTEL exam topic comparing the three switching paradigms:

Feature / Dimension	Circuit Switching	Virtual Circuit Packet Switching	Datagram Packet Switching
Path Setup Phase	Mandatory before data flow.	Mandatory (Call Request packet).	None (Immediate transmission).
Dedicated Bandwidth	Yes (Reserved channels).	No (Shared physical links).	No (Shared physical links).
Packet Routing	N/A (Physical circuit).	Fixed logical path via VCN.	Dynamic per-packet routing.
Header Content	None during data transfer.	Virtual Circuit Number (VCN).	Full Source & Destination IP.
Link Failure Impact	Connection drops completely.	Virtual circuit drops; setup required.	Reroutes automatically around failure.
Packet Arrival Order	Strictly guaranteed in order.	Guaranteed in order.	May arrive out-of-order.

3.4 The 7-Layer OSI Model Architecture

The Open Systems Interconnection (OSI) model proposed by ISO standardizes networking functions into 7 modular layers. Layering simplifies design, enables vendor interoperability, and prevents changes in one layer from affecting others.

Layer # & Name	Layer Type	Primary Technical Function	Key Protocols & Network Units
7. Application	Host-to-Host (Upper)	Network services directly to end-user applications.	HTTP, SMTP, FTP, DNS (Data)
6. Presentation	Host-to-Host (Upper)	Data syntax representation, encryption, compression, EBCDIC-ASCII conversion.	TLS/SSL, JPEG, ASCII (Data)
5. Session	Host-to-Host (Upper)	Establishes, coordinates, and terminates application conversations/logins.	NetBIOS, RPC, Sockets (Data)
4. Transport	Host-to-Host (Upper)	End-to-end reliable data delivery, error recovery, flow control.	TCP, UDP (Segment / Datagram)
3. Network	Point-to-Point (Lower)	Logical addressing, packet routing across multiple subnets.	IP, ICMP, ARP, Routers (Packet)
2. Data Link	Point-to-Point (Lower)	Node-to-node frame delivery, MAC addressing, error detection.	Ethernet, Switches, Bridges (Frame)
1. Physical	Point-to-Point (Lower)	Raw bit stream transmission over physical medium.	Cables, Hubs, Signal Volts (Bits)

EXAM POINT: Point-to-Point vs. Host-to-Host Layer Rule

Point-to-Point Layers (Layers 1-3) are active on every intermediate node/router along the path. Host-to-Host Layers (Layers 4-7) are active ONLY on the end source and destination devices!

3.5 Internetworking Devices Across OSI Layers

Hardware network devices operate at specific layers of the OSI stack:

- **Hubs (Layer 1 - Physical):** Dumb devices that receive raw bits and blindly broadcast them to all connected ports. Creates large collision domains.
- **Bridges & Layer-2 Switches (Layer 2 - Data Link):** Intelligent devices that learn MAC addresses, inspect incoming frame headers, and forward traffic only to the specific port connected to the destination host.
- **Routers & Layer-3 Switches (Layer 3 - Network):** Inspect IP headers, consult routing tables, and forward packets across different subnets/networks.

4.1 The TCP/IP 4-Layer Architecture vs. OSI Model

While the OSI model is a 7-layer theoretical standard, the **TCP/IP Protocol Stack** is the practical 4-layer suite that powers the global Internet. It originated from US military DARPA projects in the 1970s and serves as the universal language across Windows, Linux, Mac, and networking hardware.

TCP/IP Layer (4 Layers)	Corresponding OSI Layers	Primary Protocols Active	Core Functional Purpose
1. Application Layer	Application, Presentation, Session (Layers 5, 6, 7)	HTTP, SMTP, FTP, TFTP, DNS, SNMP	Handles application protocols, data formatting, encryption, and session control.
2. Transport Layer	Transport (Layer 4)	TCP, UDP	Provides process-to-process message delivery, error recovery, flow control, and port multiplexing.
3. Network Layer	Network (Layer 3)	IP, ICMP, IGMP, ARP, RARP	Handles logical addressing, packet routing across networks, and fragmentation.
4. Data Link / Physical	Data Link, Physical (Layers 1, 2)	Ethernet, Wi-Fi (802.11), Point-to-Point Protocol	Consolidates hardware NIC functions, MAC addressing, frame framing, and bit transmission.

4.2 Core Transport Protocols: TCP vs. UDP

The Transport layer offers two contrasting protocols based on application requirements:

Comparison Dimension	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Service Type	Connection-oriented, reliable, ordered stream.	Connectionless, unreliable datagram service.
Reliability Mechanism	Uses Acknowledgments (ACKs), sequence numbers, and automatic retransmissions.	No ACKs, no retransmissions, no error recovery.
Packet Ordering	Guarantees packets are delivered in exact order.	Packets may arrive out-of-order or be lost.
Header Size / Overhead	20 bytes minimum header (higher overhead).	8 bytes fixed header (minimal overhead, faster).
Ideal Use Cases	Web browsing (HTTP), File transfers (FTP), Email (SMTP), Remote shell (SSH).	Real-time voice/video streaming, Online gaming, DNS queries, TFTP.

4.3 Addressing Schemes in TCP/IP

To ensure data reaches the correct device and application, TCP/IP uses three distinct addresses at different layers:

Address Type	Bit Length	Active Layer	Primary Purpose & Uniqueness Scope
MAC Address (Hardware)	48 bits (6 bytes)	Data Link / Physical	Globally unique hardware ID burned into Network Interface Card (NIC). Used for local LAN delivery.
IP Address (Logical)	32 bits (IPv4) / 128 bits (IPv6)	Network Layer	Globally unique logical address identifying a device on the Internet. Used for routing across networks.
Port Number (Process)	16 bits (0 - 65535)	Transport Layer	Identifies a specific application or process on a host. Enables multiplexing multiple services on 1 IP.

4.4 Data Encapsulation & Networking Overhead Math

Concept of Encapsulation: As application data moves down the stack, each layer wraps higher-layer data inside its own control header (and trailer at Data Link layer). At the receiver, *decapsulation* strips headers layer by layer.

Step-by-Step Numerical Example (TFTP File Transfer):

Suppose a TFTP client sends **200 bytes** of application payload data:

- Application Layer (TFTP):** Adds a 4-byte TFTP header → $200 + 4 = 204$ bytes.
- Transport Layer (UDP):** Adds an 8-byte UDP header → $204 + 8 = 212$ bytes.
- Network Layer (IP):** Adds a 20-byte IP header → $212 + 20 = 232$ bytes.
- Data Link Layer (Ethernet):** Adds a 14-byte Ethernet header + 4-byte trailer → $232 + 14 + 4 = 250$ bytes **Total Frame Size**.

KEY CONCEPT TO REMEMBER: ENCAPSULATION OVERHEAD FORMULA & CALCULATION

Total Networking Overhead = 4 + 8 + 20 + 14 + 4 = 50 bytes

Overhead Percentage = (Total Overhead / Total Frame Size) × 100%

Overhead Percentage = (50 / 250) × 100% = 20% Overhead (Only 80% of transmitted frame is actual user data!).

MODULE 5: LECTURE 05 — TCP/IP PROTOCOL STACK (PART II): IP HEADER & SNIFFING

5.1 Anatomy of the IPv4 Header (20 to 60 Bytes)

The Internet Protocol (IP) layer handles connectionless, unreliable datagram routing across diverse networks. Every IPv4 packet carries a structured header containing 13 fields (minimum 20 bytes without options, up to 60 bytes maximum):

Field Name	Bit Length	Technical Purpose & Security Significance
Version	4 bits	Identifies IP version. For IPv4, binary value is 0100 (decimal 4).
Internet Header Length (IHL)	4 bits	Specifies header length in 32-bit words. Min value 5 (5 × 4 = 20 bytes). Max value 15 (15 × 4 = 60 bytes).
Type of Service (TOS)	8 bits	Specifies Quality of Service (QoS) requests (priority, throughput, delay).
Total Length	16 bits	Defines total size of IP datagram (Header + Payload) in bytes. Max size = 2 ¹⁶ - 1 = 65,535 bytes.
Identification	16 bits	Unique sequence ID assigned to datagram. Critical for reassembling fragmented packets at destination.
Flags	3 bits	Bit 0: Reserved (0). Bit 1: DF (Don't Fragment - drops packet if MTU exceeded). Bit 2: MF (More Fragments - set for all except last fragment).
Fragment Offset	13 bits	Specifies position of fragment relative to start of unfragmented datagram, measured in 8-byte units.
Time to Live (TTL)	8 bits	Hop counter preventing packets from looping indefinitely in routing loops. Decrement by 1 at each router; packet dropped when TTL = 0 with ICMP Time Exceeded.
Protocol	8 bits	Identifies higher-layer payload protocol. Common values: 6 = TCP, 17 = UDP, 1 = ICMP.
Header Checksum	16 bits	Error detection over IP header only using 1's complement sum. Recalculated at every router; corrupted headers dropped.
Source IP Address	32 bits	Logical IP address of original sender.
Destination IP Address	32 bits	Logical IP address of ultimate recipient.
Options & Padding	Variable	Optional features like Source Routing (sender dictates explicit router path) or Record Route. Padded to 32-bit boundary.

5.2 TTL and Routing Loop Prevention

Why TTL is Vital: If a network routing table contains a loop (e.g., Router A forwards to Router B, which forwards back to Router A), packets could circulate infinitely, consuming bandwidth and crashing routers.

How TTL Works Step-by-Step:

1. Sender sets initial TTL value (e.g., 64 or 128).
2. Every router ('hop') that receives the packet decrements TTL by 1.
3. Because TTL changes, the router recalculates the **Header Checksum**.
4. If TTL reaches 0, the router discards the packet and sends an ICMP 'Time Exceeded' message back to sender.

REAL-WORLD EXAMPLE / ANALOGY: TRACEROUTE TOOL WORKING PRINCIPLE

The **tracert** utility uses TTL to discover path routers! It sends packets with TTL=1 (discovers hop 1), then TTL=2 (discovers hop 2), and so on, using ICMP Time Exceeded error responses to map every router on the route.

5.3 Packet Sniffing and Traffic Analysis (Wireshark in Labs)

What it is: A packet sniffer (such as Wireshark) is a software tool that captures live network frames passing through a Network Interface Card (NIC) and displays protocol headers layer-by-layer.

Educational Lab Usage: Ethical hackers and network administrators use sniffers in authorized lab environments to: (1) Verify protocol encapsulation, (2) Diagnose low TTL or corrupted checksum issues, (3) Detect unencrypted passwords transmitted over plain HTTP/FTP, and (4) Analyze malicious traffic patterns.

MODULE 6: WEEK 1 REVISION & NPTEL EXAM PREPARATION PAGES

- **Ethical Hacking:** Authorized proactive security testing using malicious hacker tools to discover and patch flaws under strict legal permission.
- **Penetration Models:** White Box (full info), Black Box (zero info), Gray Box (partial info).
- **Network Types:** LAN (local, Ethernet up to 100Gbps+), WAN (large scale, leased/provider lines), MAN (city scale).
- **Switching:** Circuit Switching (dedicated path, guaranteed bandwidth, inefficient for bursty data), Packet Switching (segmented packets, shared links, store-and-forward buffering), Virtual Circuit (connection-oriented packet switching with static path and VCN).
- **Datagram Approach:** Independent packet routing, no call setup, highly resilient, but causes out-of-order delivery, packet loss, and duplicates.
- **Delays:** Propagation Delay (distance/speed), Transmission Time (message length/bandwidth), Processing & Queueing delays.
- **OSI 7 Layers:** Physical, Data Link, Network, Transport, Session, Presentation, Application. Lower 3 = Point-to-Point; Upper 4 = Host-to-Host.
- **TCP/IP 4 Layers:** Application (5,6,7), Transport (4), Network (3), Data Link/Physical (1,2).
- **Addressing:** MAC (48-bit hardware, LAN), IP (32-bit IPv4 / 128-bit IPv6, global routing), Port (16-bit, process multiplexing).
- **Protocols:** TCP (reliable, connection-oriented, ACKs) vs UDP (connectionless, unreliable, lightweight).
- **IP Header:** Min 20 bytes. Key fields: IHL, Total Length, Identification, Flags (DF, MF), Fragment Offset, TTL (loop prevention), Protocol, Header Checksum.

2. Master Formula Sheet

All mathematical formulas required for networking calculations in Week 1:

Formula Name	Mathematical Equation	Variable Definitions & Units	Calculation Purpose
Propagation Delay	$T_{prop} = d / s$	d = Distance (meters) s = Speed of light in medium (m/s)	Calculates time taken for a signal to travel across physical distance.
Transmission Time	$T_{trans} = L / R$	L = Packet Size (bits) R = Link Bandwidth / Rate (bps)	Calculates time required to push all packet bits onto medium.
Total Network Delay	$T_{total} = T_{prop} + T_{trans} + T_{proc} + T_{queue}$	T_{proc} = Processing delay T_{queue} = Queueing delay	Computes total end-to-end latency across network path.
IP Header Length	Header Bytes = IHL $\times$ 4	IHL = 4-bit Header Length field (5 to 15)	Converts IHL word count into total IP header size in bytes.
Encapsulation Overhead	Overhead % = (Header Bytes / Total Frame) $\times$ 100%	Header Bytes = Sum of headers/trailers Total Frame = Payload + Headers	Calculates percentage of transmitted frame consumed by networking headers.

3. Important Solved Numerical Examples

SOLVED EXAMPLE 1: Propagation Delay & Transmission Time

Problem: A host sends a **1,000-byte** packet over a **1,000 km** fiber optic link with a bandwidth of **1 Gbps** (1,000,000,000 bps). Propagation speed in fiber is **2×10^8 m/s**. Calculate (a) Propagation Delay and (b) Transmission Time.

Step-by-Step Solution:

- **Convert units:** $d = 1,000 \text{ km} = 1,000,000 \text{ meters}$. $L = 1,000 \text{ bytes} \times 8 = 8,000 \text{ bits}$. $R = 10^9 \text{ bps}$.
- **(a) Propagation Delay:** $T_{prop} = d / s = 1,000,000 / (2 \times 10^8) = 0.005 \text{ seconds} = \mathbf{5 \text{ milliseconds (ms)}}$.
- **(b) Transmission Time:** $T_{trans} = L / R = 8,000 / 10^9 = 0.000008 \text{ seconds} = \mathbf{0.008 \text{ ms (8 microseconds)}}$.
- **Final Answer:** $T_{prop} = 5 \text{ ms}$, $T_{trans} = 0.008 \text{ ms}$. (Notice T_{prop} dominates long distance links!).

Problem: A packet sniffer captures an IPv4 datagram with an IHL value of 5 and a Total Length value of 604. What is the size of the IP header and the size of the data payload?

Step-by-Step Solution:

- **Header Size:** Header Bytes = $IHL \times 4 = 5 \times 4 = 20$ bytes.
- **Payload Size:** Payload = Total Length - Header Bytes = $604 - 20 = 584$ bytes.
- **Final Answer:** IP Header = 20 bytes, Data Payload = 584 bytes.

4. NPTEL Exam Practice Questions (MCQs)

Q1: Which penetration testing model provides the tester with full information about target infrastructure?

A) Black Box Model | B) White Box Model | C) Gray Box Model | D) Red Box Model

KEY CONCEPT TO REMEMBER: ANSWER & EXPLANATION

Correct Answer: B | Explanation: White Box provides complete topology, IP schemes, and credentials for thorough testing.

Q2: What is the main disadvantage of Circuit Switching for computer data traffic?

A) High queuing delays | B) Out-of-order packet arrival | C) Inefficient bandwidth utilization for bursty traffic | D) Lack of guaranteed bandwidth

KEY CONCEPT TO REMEMBER: ANSWER & EXPLANATION

Correct Answer: C | Explanation: Circuit switching reserves dedicated channels during idle silent gaps in bursty data.

Q3: Which component of network delay depends directly on the physical medium speed and distance?

A) Transmission Time | B) Queueing Delay | C) Propagation Delay | D) Processing Delay

KEY CONCEPT TO REMEMBER: ANSWER & EXPLANATION

Correct Answer: C | Explanation: Propagation delay $T_{prop} = d / s$ depends on distance d and medium propagation speed s .

Q4: At which OSI layer do Routers and Layer-3 Switches operate?

A) Data Link Layer (Layer 2) | B) Network Layer (Layer 3) | C) Transport Layer (Layer 4) | D) Application Layer (Layer 7)

KEY CONCEPT TO REMEMBER: ANSWER & EXPLANATION

Correct Answer: B | Explanation: Routers make forwarding decisions based on IP addresses at Layer 3 (Network Layer).

Q5: What is the primary purpose of the Time to Live (TTL) field in an IPv4 header?

A) Error detection in header | B) Reassembling fragments | C) Preventing infinite routing loops | D) Identifying transport protocol

KEY CONCEPT TO REMEMBER: ANSWER & EXPLANATION

Correct Answer: C | Explanation: TTL decrements by 1 at each router hop; dropped at 0 to prevent infinite routing loops.

Q6: Which protocol provides a connectionless, fast, lightweight transport service without ACKs?

A) TCP | B) UDP | C) IP | D) ARP

KEY CONCEPT TO REMEMBER: ANSWER & EXPLANATION

Correct Answer: B | Explanation: UDP is a connectionless transport protocol with low 8-byte overhead and no retransmission ACKs.

Q7: In an IPv4 header, if the IHL field contains decimal 7, what is the IP header size in bytes?

A) 7 bytes | B) 20 bytes | C) 28 bytes | D) 35 bytes

KEY CONCEPT TO REMEMBER: ANSWER & EXPLANATION

Correct Answer: C | Explanation: Header size in bytes = $IHL \times 4 = 7 \times 4 = 28$ bytes.

A) RARP | B) ICMP | C) ARP | D) IGMP

KEY CONCEPT TO REMEMBER: ANSWER & EXPLANATION

Correct Answer: C | Explanation: Address Resolution Protocol (ARP) maps a target IP address to its physical MAC address.

5. Viva Voce / Oral Exam Questions

V1: What is the fundamental difference between Ethical Hacking and Cracking?

Answer: Ethical hacking is authorized security testing performed with legal permission to strengthen defenses. Cracking is unauthorized security breaching conducted maliciously.

V2: Why is store-and-forward buffering important in packet-switched routers?

Answer: Store-and-forward allows routers to receive complete packets, verify checksums, adapt between fast incoming and slow outgoing links, and prioritize urgent traffic.

V3: Differentiate between Point-to-Point layers and Host-to-Host layers in the OSI model.

Answer: Point-to-Point layers (Layers 1-3) are active on all intermediate routers along the path. Host-to-Host layers (Layers 4-7) are active only at source and destination end hosts.

V4: What happens when an IPv4 packet's TTL reaches zero at a router?

Answer: The router drops the packet and sends an ICMP 'Time Exceeded' error message back to the source host.

V5: Explain why packet encapsulation creates networking overhead.

Answer: Each stack layer appends its own control header (IP, UDP, Ethernet headers) to payload data. Total bytes sent exceed user payload; $\text{Overhead \%} = (\text{Header Bytes} / \text{Total Frame}) \times 100\%$.

6. Final Exam Readiness Checklist

Check off each item once you have mastered the concept:

- ☐ I can explain the legal difference between Ethical Hacking and Cracking.
- ☐ I know the information levels of White Box, Black Box, and Gray Box testing models.
- ☐ I can explain why Circuit Switching is inefficient for bursty computer data.
- ☐ I know the 3 mechanisms of Store and Forward buffering in routers.
- ☐ I can solve Propagation Delay ($T_{\text{prop}} = d/s$) and Transmission Time ($T_{\text{trans}} = L/R$) problems.
- ☐ I can list all 7 layers of the OSI model and distinguish Point-to-Point vs Host-to-Host layers.
- ☐ I know the 4 layers of TCP/IP and their corresponding OSI layers.
- ☐ I can compare TCP and UDP across reliability, overhead, and use cases.
- ☐ I can calculate IHL header bytes ($\text{IHL} \times 4$) and encapsulation overhead percentage.
- ☐ I understand how the TTL field prevents infinite routing loops in IP networks.

KEY CONCEPT TO REMEMBER: END OF WEEK 1 STUDY NOTEBOOK

Congratulations on completing Week 1 study notes! Review all numerical examples, comparison tables, and MCQs prior to your NPTEL examination.